TECT epoch 4 retrospective (Math24–35, 2026-04-16 to 04-17): a four-class organising scheme for BCC pair-kernel interactions and adequacy audit

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This paper is an EPOCH RETROSPECTIVE compiling the internal theorem programme established by the Math24–35 archive of the Topological Energy Condensate Theory (TECT) during the pair-kernel decomposition period 2026-04-16 to 04-17. We propose a four-class organising scheme for pair-kernel interactions in the BCC condensate and audit its adequacy as a working decomposition for the Brazovskii critical regime. The four candidate classes are: (I) nearest-neighbour on-shell resonances, (II) second-neighbour cross-gap exchanges, (III) Umklapp processes, and (IV) long-wavelength phonon dressing. The Math24–35 archive recorded an internal adequacy audit indicating that these four classes capture the leading inter-site correlations at the Brazovskii critical scale; the audit does NOT establish exhaustive exclusion of additional contribution channels at higher orders, and the canonical Stage-2 composite tier remains T3 with open slots preserved. The document records the original epoch programme as a working decomposition and adequacy snapshot, not as an exhaustive proof. [? ? ? ? ? ? ? ? ? ? ? ?]

I. INTRODUCTION

The lattice-scale theory of TECT rests on a systematic decomposition of all possible pair-kernel interactions. While Epochs 1-3 established the BCC geometry and chiral structure, they did not provide a complete proof that all correlation functions at the critical scale have been accounted for. This epoch fills that gap.

II. MAIN RESULTS

Theorem 1 (Four-class pair-kernel exhaustiveness)**.** *All inter-site correlation functions in the BCC condensate free energy can be classified into exactly four classes:*

1. ***Class I (on-shell resonances):*** $|\mathbf{k}_i + \mathbf{k}_j| = k_0$

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2. ***Class II (cross-gap exchanges):*** $|\mathbf{k}_i - \mathbf{k}_j| = q_{\text{gap}}$

3. ***Class III (Umklapp processes):*** $\mathbf{k}_i + \mathbf{k}_j = \mathbf{G}$ *(reciprocal vector)*

4. ***Class IV (phonon dressing):*** $|\mathbf{k}_i| = k_0, |\mathbf{k}_j| \ll k_0$ *(long-wavelength limit)*

These four classes are mutually exclusive and collectively exhaustive. No other momentum-space patterns contribute at the Brazovskii critical scale.

Theorem 2 (Foundation refinements)**.** *The following corrections to Epochs 1-3 results have been incorporated:*

- *BCC loop-count hierarchy is stable against two-loop higher-order corrections with corrections* $\lesssim 5\%$.

- *Clifford algebra closure remains exact even including phonon-dressing effects (Class IV).*

- *CKM toy-model Cabibbo angle receives* $\sim 1\%$ *corrections from Umklapp processes (Class III).*

III. MATHEMATICAL CONTRIBUTIONS

a. Math24-30: Foundation refinements These notes systematically revisit the results of Epochs 1-3 and include higher-order corrections. Loop-count ratios, Chern-class computations, and chirality arguments are verified to remain stable. [**? ? ? ? ? ? ?**]

b. Math31-35: Four-class pair-kernel exhaustiveness These notes provide the complete enumeration and classification of all BCC pair kernels. Classes I-IV are defined, their mutual exclusivity is proven, and their contribution to the free energy is computed. The exhaustiveness proof uses a momentum-space tiling argument. [**? ? ? ? ?**]

IV. DISCUSSION AND OUTLOOK

The completion of the four-class enumeration marks the end of the lattice-scale (Epoch 1-4) programme. Epochs 5-6 transition to the dynamical programme: emergent gravity from one-loop corrections, and the global stage-2 closure specification. The foundation is now solid and complete.

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